L04: Linear Regression: Optimization and Regularization

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Recap:

- Problems of interest:
  - Learn a function $y = f(\mathbf{W}, \mathbf{x})$ from the data.
    - (a) Classification (b) Regression
  - Learn Feature Transformations $\mathbf{x}' = \mathbf{W}\mathbf{x}$ or $\mathbf{x}' = f(\mathbf{W}, \mathbf{x})$
- Algorithms/Approaches:
  - Nearest Neighbour Algorithm
  - Linear Classification: $sign(\mathbf{w}^T\mathbf{x})$
  - Decide as ω_1 if $P(\omega_1|\mathbf{x}) \geq P(\omega_2|\mathbf{x})$ else ω_2.
  - Linear Regression
- Performance Metrics:
  - Accuracy, TP/FP etc., Confusion Matrix; Ranking: Precision, Recall, F-Score, AP
- Supervised Learning:
  - Notion of Training, Validation and Testing
  - Notion of Loss Function, MSE, Regularization (today)
  - Role of Optimization, Convex and non-Convex optimization
  - Closed form solution, Gradient Descent, Eigen vector solutions (today)

Linear Regression

Given N examples $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}$. Our goal is to find a solution of the form: $y_i = mx_i + c$; or $y_i = w_1 x_i + w_2$ or

$$\hat{y}_i = \mathbf{w}^T \mathbf{x}_i$$

Problem:

$$\mathbf{w}^* = \arg\min_{\mathbf{w}} \sum_{i=1}^{N} (y_i - \mathbf{w}^T \mathbf{x}_i)^2$$

Equivalent problem

$$\mathbf{w}^* = \arg\min_{\mathbf{w}} ||\mathbf{y} - \mathbf{X}\mathbf{w}||_2^2$$

[1]

[1] L_p norm of a vector $\mathbf{x} = (\sum_{i=1}^{d} |x_i|^p)^{\frac{1}{p}}$

Two Solutions

1. Closed Form Solution

$$\mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}$$

2. Gradient Descent Solution

$$\mathbf{w}^{k+1} \leftarrow \mathbf{w}^k - \eta\nabla J$$

$$\min_{\mathbf{w}} J(\mathbf{w}) = \min_{\mathbf{w}} \frac{1}{N}\sum_{i=1}^{N}(y_i - \mathbf{w}^{\mathbf{T}}\mathbf{x_i})^2$$

$$\nabla J = \frac{2}{N}\sum_{i=1}^{N}(y_i - \mathbf{w}^{\mathbf{T}}\mathbf{x_i})(-\mathbf{x_i})$$

Regularization for Optimization

- Prefer some type of solutions (eg. prefer sparse solutions) over others.
- Frequently used in solving ill-posed problems (eg. avoid singularities such as inverting a near singular matrix)
- A set of methods for reducing overfitting in machine learning models.
- In short, critical for managing model complexity, improving generalization to new data, and addressing many specific issues
- Example: Add an additional function to the loss function

$$J = f(\mathbf{w}, \mathbf{x}, y) + \lambda g(\mathbf{w})$$

Regularize with L0, L1 and L2

$$J = f(\mathbf{w}) + \lambda||\mathbf{w}||$$

It is common to regularize with $L2$, $L1$ (ideally $L0$; why?) norms.
Recap: L_p norm of of $\mathbf{x}$ is defined as

$$||\mathbf{x}|| = (\sum_{i=1}^{d} |x_i|^p)^{\frac{1}{p}}$$

- Em. L1: (when d=2): $||\mathbf{x}||_1 = |x_1| + |x_2|$
- Eg. L2: (when d=2): $||\mathbf{x}||_2 = \sqrt{x_1^2 + x_2^2}$
- The L0 norm of a vector counts the number of non-zero elements in the vector.
- The L_∞ infinity norm, also known as the "max norm," is a vector norm that measures the maximum absolute value of the vector elements.

Some Intuitive Arguments on Sparsity

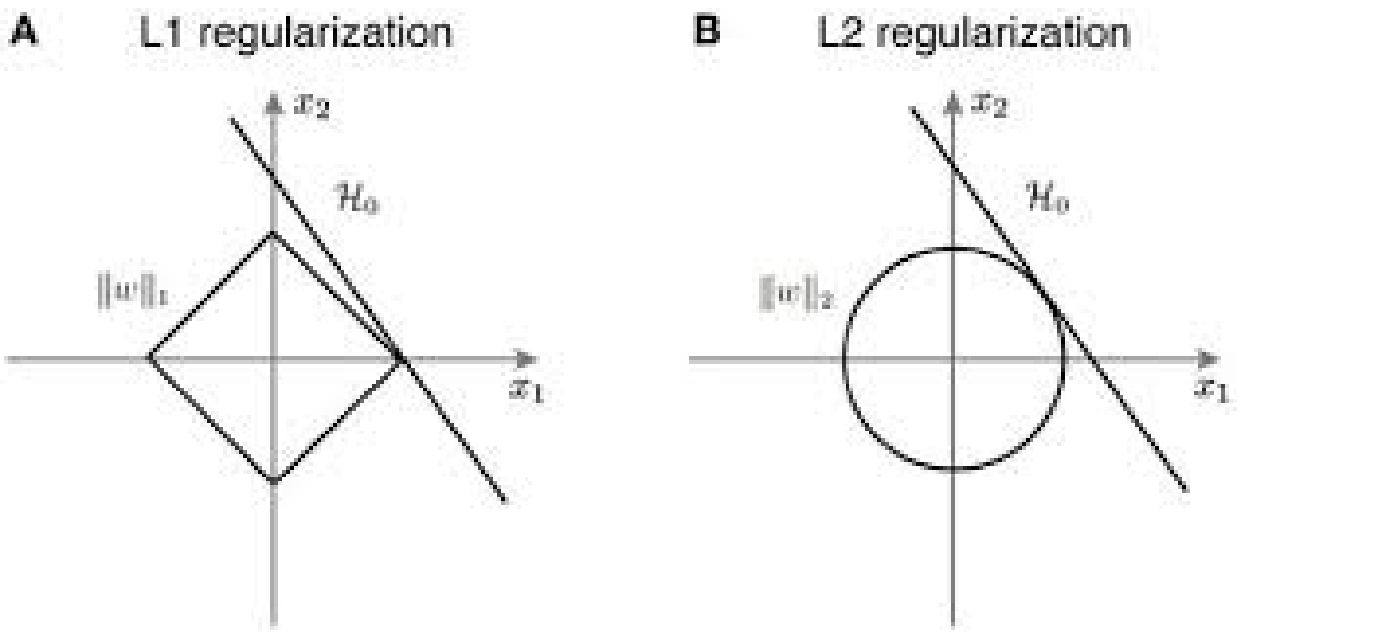
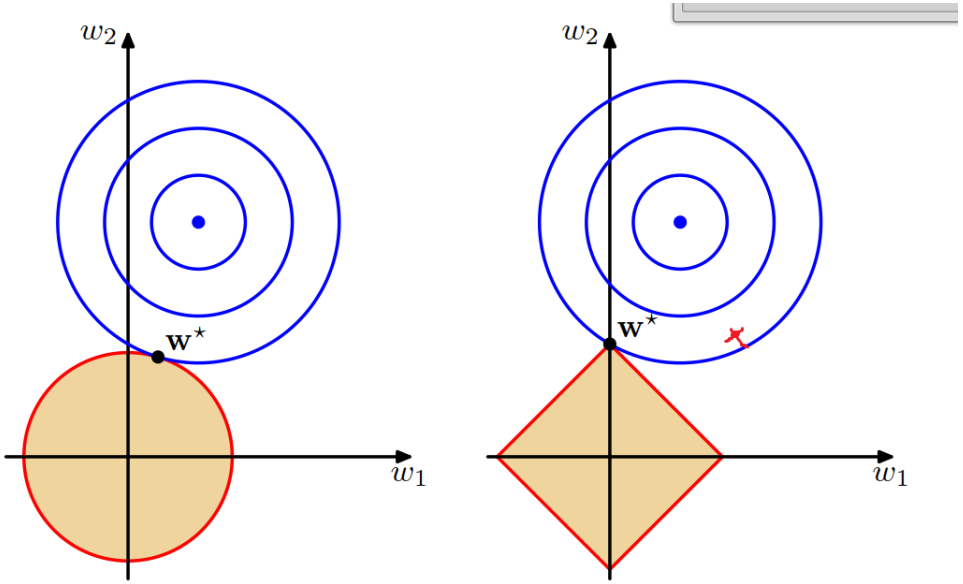
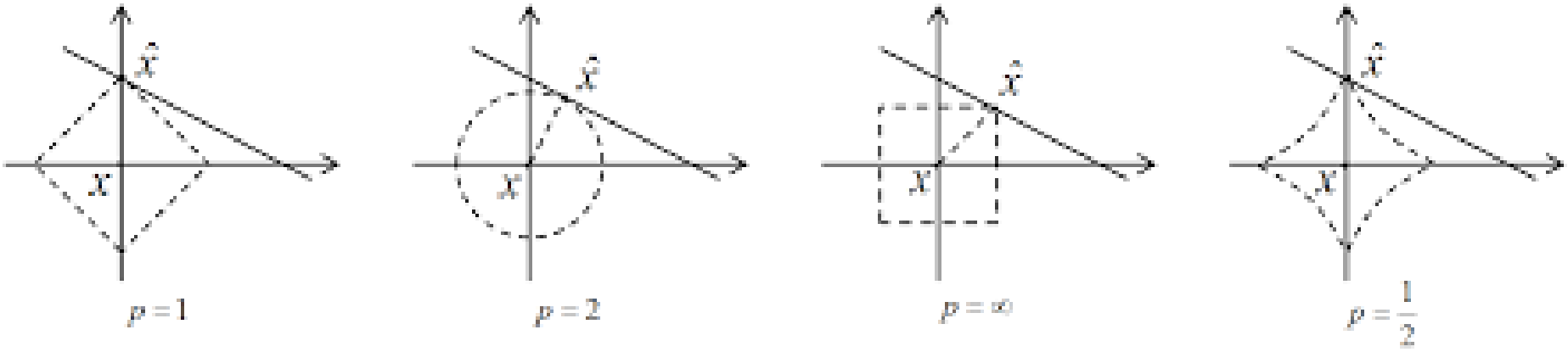


Figure 3.4 Plot of the contours of the unregularized error function (blue) along with the constraint region (3.30) for the quadratic regularizer $q = 2$ on the left and the lasso regularizer $q = 1$ on the right, in which the optimum value for the parameter vector $\mathbf{w}$ is denoted by $\mathbf{w}^\star$. The lasso gives a sparse solution in which $w_1^\star = 0$.



Some Intuitive Arguments on Sparsity



Ridge Regression

In ridge regression, we improve the objective as

$$J = \sum_{i=1}^{N} (y_i - \mathbf{w}^T \mathbf{x})^2 + \lambda \sum_{j=1}^{d} w_j^2$$
$$= \sum_{i=1}^{N} (y_i - \mathbf{w}^T \mathbf{x})^2 + \lambda ||\mathbf{w}||_2^2.$$

Let us come back to ridge regression objective. It can be rewritten as:

$$J(\mathbf{w}) = \frac{1}{N}(\mathbf{Y}^T\mathbf{Y} + \mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w} - 2\mathbf{w}^T(\mathbf{X}^\mathbf{T}\mathbf{Y}) + \lambda\mathbf{w}^T\mathbf{w}))$$

Optimization of the above objective function leads to:

$$\mathbf{w} = (\mathbf{X}^T\mathbf{X} + \lambda I)^{-1}\mathbf{X}^T\mathbf{Y}$$

- How does this help/influence in numerical stability?
- Design a gradient descent algorithm for Ridge regression
- How do we find λ?

Lasso Regression

Lasso is another variant of the regression where the objective has $L1$ norm of the $\mathbf{w}$ instead of $L2$.

$$J = \sum_{i=1}^{N} (y_i - \mathbf{w}^T \mathbf{x})^2 + \lambda \sum_{i=1}^{d} |w_j|$$

$$= \sum_{i=1}^{N} (y_i - \mathbf{w}^T \mathbf{x})^2 + \lambda ||\mathbf{w}||_1.$$

In Lasso, we not only penalize the high value of some of the coefficients, optimization also leads to zero for irrelevant variables.
The change in the norm of the penalty may seem like only a minor difference, however the behavior of the $L1$-norm is significantly different than that of the $L2$-norm.

- How do we implement Lasso regression?
- Solution to Lasso is outside the scope of this course

Recap

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

Solutions to this:

- Eigen values: $\lambda_1, \lambda_2 \ldots$
- Eigen vector: $\mathbf{v}_1, \mathbf{v}_2 \ldots$

Often λ_is are sorted in decreasing order
Often $\mathbf{v}_i$ are normalized to unit norm.

Optimization problems with Eigen Vectors as Solutions [2]

Problem: Maximize $\mathbf{w}^T\mathbf{A}\mathbf{w}$ such that $\mathbf{w}^T\mathbf{w} = 1$ (or $||\mathbf{w}|| = 1$)

We form an objective with the help of a lagrangian (λ) as

$$J(\mathbf{w}, \lambda) = \mathbf{w}^T\mathbf{A}\mathbf{w} - \lambda(\mathbf{w}^T\mathbf{w} - 1)$$

Differentiating wrt $\mathbf{w}$ and equating to zero leads to:

$$\mathbf{A}\mathbf{w} = \lambda\mathbf{w}$$

Or $\mathbf{w}$ is the eigen vector corresponding to the largest eigen value. (Why largest EV?)

[2] Benyamin Ghojogh et al, Eigenvalue and Generalized Eigenvalue Problems: Tutorial, Arxiv, 2023. An excellent read beyond this course.

Problem 1

We are interested in solving an overdetermined system of homogeneous equations

$$\mathbf{A}\mathbf{x} = \mathbf{0}$$

where $\mathbf{A}$ is $m \times n$.
Problem is formulated as:

$$\arg\min_{\mathbf{x}} ||\mathbf{A}\mathbf{x}|| \quad \text{Subject to: } ||\mathbf{x}|| = 1$$

Solution to this problem is:

- (a) Eigen Vector corresponding to the largerst eigen value of $\mathbf{A}\mathbf{A}^T$.
- (b) Eigen Vector corresponding to the largerst eigen value of $\mathbf{A}^T\mathbf{A}$.
- (c) Eigen Vector corresponding to the smallest eigen value of $\mathbf{A}\mathbf{A}^T$.
- (d) Eigen Vector corresponding to the smallest eigen value of $\mathbf{A}^T\mathbf{A}$.

Read later[3]

[3] https://cw.fel.cvut.cz/old/_media/courses/a3m33iro/overdetermined_homogenous_linear_equations.pdf

Problem 2

Consider a square matrix $\mathbf{A}$ with eigen values λ_i and eigen vectors $\mathbf{v}_i$. Then for $\mathbf{A}^T$,

- (a) Eigen values and eigen vectors are the same as that of $\mathbf{A}$.
- (b) Eigen values are the same. Eigen vectors are $\mathbf{v}^T$.
- (c) Eigen values are $\frac{1}{\lambda_i}$
- (d) We can comment for symmetric matrix $\mathbf{A}$. But not for other square matrices.
- (e) None of the above.

The the eigen values of $\mathbf{A}$ are λ_i, what are the eigen values of $\alpha\mathbf{A}$, where α is a scalet.

- (a) λ_i itself.
- (b) $\frac{\alpha}{\lambda_i}$
- (c) $\alpha\lambda_i$
- (d) $\frac{1}{\alpha}\lambda_i$
- (e) None of the above.

Discussion Point - I

We are worried about outliers in the regression. Let us give a score γ_i as the "importance" or "confidence" of a sample that this is an inlier. We can now modify the loss/objective as:

$$\sum_{i=1}^{N} \gamma_i \times (y_i - \mathbf{w}^T \mathbf{x}_i)^2$$

- Write down the objective in matrix form. (Hint: use $\Gamma = Diag(\gamma_i)$). What is the final closed form expression for the $\mathbf{w}$?
- (Advanced) Consider a two step Itertive algorithm:
  - Assign γ_i as inversely proportional to the distance from the line. (distance $=0 \rightarrow \gamma$ as 1 and high distance $\rightarrow \gamma$ as 0)
  - Compute $\mathbf{w}$ using the closed form expression (Q1).

  If we iterate the above two steps? (i) will it converge? (ii) will it take care of outliers? (Later: Try it out on a toy data of yours [4])

[4] A similar treatment in a different area: read "Sample weighted Clustering Methods", CMA, 2011

Discussion Point - II

(use notations and conventions from the class) Consider the problem of linear regression where we minimize the loss

$$\mathcal{L}_1 = \frac{1}{N}\sum_{i=1}^{N} \alpha_i (y_i - \mathbf{w}^T \mathbf{x}_i)^2 + \lambda_1 g(\mathbf{w})$$

where $g()$ is a regularization term. We also write the loss in matrix form as

$$\mathcal{L}_2 = \frac{1}{N}[Y - \mathbf{X}\mathbf{w}]^T A [Y - \mathbf{X}\mathbf{w}] + \lambda_2 g(\mathbf{w}).$$

- What can we say about the matrix A? size? properties? elements?
- What can we say about the optima, equivalence?
- What can we say when the objective is regularized?
- When regularizing what happens to the objective/optimal solution?

Review Question - I (one, none or more correct)

Consider a square matrix $\mathbf{A}$ with eigen values λ_i and eigen vectors $\mathbf{v}_i$. Then for $\mathbf{A}^T$,

- (a) Eigen values and eigen vectors are the same as that of $\mathbf{A}$.
- (b) Eigen values are the same. Eigen vectors are $\mathbf{v}^T$.
- (c) Eigen values are $\frac{1}{\lambda_i}$
- (d) We can comment for symmetric matrix $\mathbf{A}$. But not for other square matrices.
- (e) None of the above.

Review Question - III (one, none or more correct)

Consider a data matrix $\mathbf{X}$ where every column is mean subtracted samples. We compute $\mathbf{A} = \mathbf{X}\mathbf{X}^{\mathbf{T}}$. Then $\mathbf{A}$

- is a $d \times d$ matrix
- is a $N \times N$ matrix
- is a symmetric matrix
- is a scaled version of the covariance matrix
- none of the above

Review Question - IV (one, none or more correct)

The inverse of an upper triangular matrix is:

- (a) Upper triangular
- (b) Lower triangular
- (c) Need not be triangular.
- (c) Will not exist
- (d) None of the above

Review Question - V (one, none or more correct)

The the eigen values of $\mathbf{A}$ are λ_i, what are the eigen values of $\alpha\mathbf{A}$, where α is a scalet.

- (a) λ_i itself.
- (b) $\frac{\alpha}{\lambda_i}$
- (c) $\alpha\lambda_i$
- (d) $\frac{1}{\alpha}\lambda_i$
- (e) None of the above.